



**KUALA LUMPUR UNIVERSITY OF  
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| <b>CW2:</b>                                                                                                                                                                                                                                                                                                                              | Individual Research Assignment                                                                                              |
| <b>Assessment number<br/>&amp;title</b>                                                                                                                                                                                                                                                                                                  | <b>Comparison of Excel and Power BI for Business Reporting</b><br><br><i>A Practical Evaluation Using Retail Sales Data</i> |
| <p>I declare that all material in this assignment/project/tutorial is my own work except where there is clear acknowledgement or reference to the work of others, and I have complied with and agreed to the University statement on Plagiarism and Academic Integrity.</p> <p>Signature: <i>Mohamed</i> Date: 5/9/2026</p> <p>.....</p> |                                                                                                                             |

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# 1. Introduction

Data is collected by every single company. However, data alone cannot be of use to anybody. It will become meaningful when somebody collects it, cleans it, examines it, and then translates it into something that is actionable by others (Chen et al., 2012). That is where Business Analytics Tools come in. There are two of the most commonly utilized Business Analytics tools today: Microsoft Excel and Microsoft Power BI, especially in Small and Medium-Sized Enterprises (SMEs), which generally lack large IT teams (Scholz et al., 2010). While both tools are developed by Microsoft, they are designed for entirely different functions, and selecting the correct tool could greatly reduce the amount of time and cost associated with implementing either one. This document provides a comparative analysis of Excel and Power BI as Business Reporting tools. Rather than only discussing these tools theoretically, this document is based upon a real-world, hands-on test utilizing a real-world dataset, the Sample Superstore Dataset from Kaggle. This document will detail how each tool processes collecting the data, cleaning the data, analyzing the data, visually presenting the data, and utilizing the data for Business Reporting. In addition to detailing how each tool performs these functions, the document will highlight the advantages and disadvantages of each tool in a Malaysian SME Retailer context to ensure that the comparison remains relevant to a realistic/practical scenario instead of being solely academic.

## 2. Overview of the Selected Tools

Microsoft Excel has been used by businesses as one of the primary tools for creating and using data, building financial models, and performing quick analyses since the late 1980s (Chen et al., 2012). The greatest advantage of Microsoft Excel lies in its ability to be flexible. Users can create formulas, build PivotTables, and create charts around raw data quickly; further, many working adults have already learned to use Excel at least at a basic level. Also, when a company purchases Microsoft Office, they do not need to purchase additional software to use Excel. However, Power BI is different. It was developed by Microsoft in 2011 as a Business Intelligence (BI) Tool designed to allow individuals with no formal technical training or experience to build their own interactive reports and dashboards. This means that an individual would not require a BI Team Specialist to assist him/her in developing these types of reports. There are three components to Power BI: Desktop App - allows users to develop reports; Online Service - provides users with the capability to share the reports; Mobile App - allows users to

view the reports. Additionally, Power BI uses the same type of engine and data model as other Microsoft products such as SQL Server and SharePoint. Therefore, Power BI is capable of handling larger amounts of data and providing greater levels of complexity than standard Excel files (Microsoft, 2024). Since Power BI is also produced by Microsoft, an individual who already understands PivotTables within Excel should find the concept of Power BI's charts and fields relatively easy to understand. To put it simply, Excel is a spreadsheet with some analysis capabilities, whereas Power BI is a product developed specifically for Business Intelligence, and each reflects the above-described differences throughout all stages.

### **3. Data Gathering and Import Process**

The same data was utilized in testing both tools; the Sample Superstore dataset, an actual dataset that can be accessed at Kaggle, has 9,994 records and includes 21 fields such as: Order Date, Region, Category, Sub-Category, Sales, Quantity, Discount, and Profit. The Sample Superstore dataset was narrowed down by filtering the Central region orders from 2017 to give a subset size of approximately 780 rows. The CSV file was first opened in Excel, where it automatically created a Table. The Region and Order Date fields were then filtered to produce the smaller working set (See Figure 1). Afterward, the same filtered file was loaded into Power BI using Get Data > Text/CSV. It was determined whether the correct sheet had been selected before loading it using the Navigator window (See Figure 2). As this step demonstrates, there are differences in how each tool performs this function. Importing into Excel involves manually importing a single file. While importing a single file will work well for one individual who will use only one file, getting data in Power BI utilizes the "Get Data" tool to regularly import data from multiple data types (i.e., database or cloud) and allows users to repeat imports as needed. This is another explanation why self-service business intelligence tools have become so prevalent in organizations conducting ongoing reporting needs (Ramakrishnan et al., 2012).

### **4. Data Cleaning and Preparation**

Business Intelligence (BI) effectiveness is heavily dependent upon data quality. Data collection and preparation are key drivers of success for any BI implementation (Ramakrishnan et al., 2012). Rather than simply assuming the provided data contained errors, this report examined the working dataset for accuracy within both BI tools. In Excel, "Remove Duplicates" was utilized to identify duplicate rows in the entire working dataset, which included 21 columns.

No duplicates were identified (see Figure 3). The report then used a COUNTIFS formula to determine if there were multiple instances of an identical Order ID and Product ID combination with differing quantity values. As shown in Figure 4, no such cases existed. Within Power BI, the exact same file was analyzed within the Power Query Editor. Remove Duplicates was once again run, but the data types associated with each column were also reviewed to verify that Order Date was recognized as a date field; Sales and Profit fields were verified to be numeric rather than text. Figures 3 and 4 show that neither tool indicated any errors or issues with the data quality in the working dataset. Even though this provides a positive conclusion, it does provide value. The two tools demonstrated the capability to allow users to effectively validate data quality regardless of whether their findings indicate "there are no problems", since validation of data quality is always an important aspect of proper analytical practice.

## 5. Analysis and Visualisation Functions

In Excel, I used PivotTables to create summaries for Sales and Profit for each Category and Sub-Category. Additionally, I created separate PivotTables for Sales by month, based upon the Order Date field. For both tables, I converted them to corresponding PivotCharts. As such, I have a bar chart that shows Sales and Profit by category (see Fig. 6) and a line chart showing the Sales trend throughout 2017 (see Fig. 7).

The same type of thing was done in Power BI, however, instead of static charts, they are interactive. There are three sections on the main dashboard page: A Card displaying the total profit, a line chart of Sales by month, and a bar chart that compares Sales and Profit among the three categories (see Fig. 8). Most good dashboards contain one or two large summary metrics along with a few smaller supporting charts. Therefore, users should be able to move from the larger view down to specifics very quickly (Few, 2013), as was accomplished with the Power BI dashboard.

One interesting discovery was made during this phase; while there were high sales within the Furniture category, the Profit bar for it was approximately \$0.00 below zero. It is therefore logical to assume that the Furniture category was generating a loss due to discounting practices. Visual comparisons of data points like these are much simpler to recognize than single metrics themselves.

## 6. Practical Business Application

The primary purpose of Business Analytics (BA) is to support better-informed business decision-making, rather than simply creating attractive visualizations (Chen et al., 2012). There were two very obvious recommendations for a Malaysian Small/Medium Enterprise (SME) operating a retail store similar to the store represented by the data used in this case study. The first recommendation was based upon the Monthly Sales Trend, which showed a consistent seasonal variation in sales. As there were distinct patterns of sales during quieter times (February and April) and busier times (June and December), a retailer could utilize this knowledge to adjust their staffing and inventory management practices accordingly. An example would include hiring additional part-time employees during the busier time frames of the month/quarter/year, thus allowing the retailer to save money without negatively affecting customer service. The second recommendation was based upon the fact that even though Furniture had been selling well, it was marginally unprofitable due to excessive discounts being applied. Therefore, a key area that the retailer will need to examine further is its Discount Policy for the specific product line of Furniture. Discounts represent a significant factor when determining whether profit can become negative. A possible way for the retailer to address this issue would be to establish a minimum discount threshold for Furniture products, thereby preventing Sales Staff from approving discounts that would result in a sale becoming unprofitable. Regarding which analytical tool(s) to use, previous studies have shown that Small/Medium Enterprises (SMEs) typically do not possess the financial means nor technical expertise required to implement full-scale Business Intelligence (BI) solutions (Scholz et al., 2010; Maroufkhani et al., 2023). Given these limitations, researchers suggest utilizing Microsoft Excel as an initial platform for BA initiatives within SME environments. Excel offers a low-cost solution that requires minimal new software installation and/or employee training, whereas Microsoft's Power BI is recommended to be utilized once the organization requires regular, interactive, and shareable reporting across multiple users (i.e., a monthly dashboard accessible to multiple Branch Managers).

## 7. Strengths and Limitations

Excel is highly adaptable; its major strengths include that nearly every user already possesses the program and has at least a basic understanding of it. This reduces the amount of time required for users to become proficient. The majority of users who have used Excel will find

the transition to using the PivotTable feature as well as creating their own custom formulas easy. Excel's greatest weakness based upon this project's findings includes being limited to use as a single-user application. Users cannot easily collaborate on files or projects due to the inability to share an updated file with multiple users unless they send the file via email or save it to a shared drive. Additionally, when working with large datasets within Excel (i.e., over hundreds of thousands of rows), performance slows significantly. During this project, one of the risks identified was related to saving a file in CSV format rather than the standard format (.xlsx). Saving a file in this manner results in all PivotTables, charts, and additional sheets being deleted from the file. Since CSV is capable of holding only plain value data, these items cannot be retained.

Power BI offers greater interactivity capabilities than most reporting tools available today. For example, if you create two different charts that are displayed together on a page and then add filters to those charts, both charts will update automatically as filters are applied. Furthermore, because Power BI allows you to publish your reports through the Power BI Service, everyone viewing your reports will always view the most current version. This eliminates the issue of having to track versions of older reports.

The main advantage of using Power Query for data cleaning is that, unlike performing manual cleaning in Excel, where you perform cleaning manually and then store it somewhere else so it may be repeated later, Power Query stores and applies cleaning processes automatically anytime new data is refreshed (Microsoft, 2024).

However, Power BI requires much more training to understand than does Excel, particularly concepts such as modeling and utilizing DAX (Data Analysis Expressions) (Schlesinger & Rahman, 2016). Moreover, full sharing capabilities require an individual to purchase a Power BI Pro license, adding expense for a small company to consider (Schlesinger & Rahman, 2016).

## **8. Conclusion**

The final report compares both Excel & Power BI across each stage of the analytical process (data collection, data preparation, visualization, reporting) utilizing a small subset of unedited sample Superstore data. Neither tool stands out as clearly superior to the other; rather, they fit into different scenarios. When doing analysis quickly & alone, with high priority being placed

upon flexibility & cost, Excel is typically the best choice. However, if you are looking at developing regular interactive reports which will need to be shared among many users, especially within larger organizations requiring multiple people to have up-to-date access to the same data, then Power BI is generally the better option. For an SME considering what their future analytic strategy should include, there are several options. The least expensive way to begin creating your analytic strategy would likely be to start with Excel for your initial explorations. You could then move to Power BI as soon as your organization begins to require more frequency of reporting, greater sharing among staff, and/or much more dependency on current/real-time data.

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# 10. Appendix: Practical Evidence

The figures below correspond to the practical tool evidence referenced throughout this report.

Figure 1

The screenshot shows an Excel spreadsheet titled 'Sample - Superstore.csv'. The data table has columns: Item ID, Order ID, Ship Mode, Customer, State, Product Category, Product Name, Sales, Quantity, Discount, Profit. The data includes various products such as 'Bush Somerset Collection Bookcase', 'Hoselink 1000 Series Steel Ring Binders', 'Hoselink 1000 Series Steel Ring Binders', 'Hoselink 1000 Series Steel Ring Binders', etc.

Figure 2

The screenshot shows a Power BI report titled 'Sample - SuperStore'. The data table has columns: Order ID, Ship Date, Ship Mode, Customer ID, Customer Name. The data includes various customers such as 'Matt Abela', 'Linda Cazzini', 'Tracy Blumst', 'Ken Brennan', 'Christopher S', 'Paul Gonzale', 'Paul Gonzale', 'Sandra Glass', 'Shirley Jacob', 'Logan Currie', 'Joseph Holt', 'Joseph Holt', 'Joseph Holt', 'Joseph Holt', 'Joseph Holt', 'Doris liebe', 'Jonathan Hov', 'Guy Thorntor', 'Guy Thorntor', 'Janet Molinar', 'Janet Molinar', 'Ken Heide', 'Brenda Bowm', 'Troy Blackwe', 'David Smith'.

Figure 3

The screenshot shows an Excel spreadsheet with a large table of data. A 'Remove Duplicates' dialog box is open, allowing the user to select which columns to use for identifying duplicates. The columns listed in the dialog are: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Post Code, Region, Product ID, Category, Sub-Category, and Product Name. The 'My data has headers' checkbox is checked.

Figure 4

The screenshot shows a pivot table summarizing sales data. The pivot table has columns for Sales, Quantity, Discount, Profit, and a duplicate check column. The data is grouped by Section, x6-15/16d x11-1/4h, and Cordless Telephone.

|                           | Sales   | Quantity | Discount | Profit    | duplicate check |
|---------------------------|---------|----------|----------|-----------|-----------------|
| Section, 5 1/2" x 3 3/16" | 29.472  | 3        | 0.2      | 9.9468    | 1               |
|                           | 147.168 | 4        | 0.2      | 16.5564   | 1               |
|                           | 19.05   | 3        | 0        | 8.763     | 1               |
|                           | 1.248   | 3        | 0.8      | -1.9344   | 1               |
|                           | 9.708   | 3        | 0.6      | -5.8248   | 1               |
| x6-15/16d x11-1/4h        | 27.24   | 3        | 0.2      | 2.724     | 1               |
|                           | 230.376 | 3        | 0.2      | -48.9549  | 1               |
|                           | 19.99   | 1        | 0        | 6.7966    | 1               |
|                           | 6.16    | 2        | 0        | 2.9568    | 1               |
|                           | 839.43  | 3        | 0        | 218.2518  | 1               |
|                           | 97.264  | 4        | 0.8      | -243.16   | 1               |
|                           | 66.284  | 2        | 0.8      | -178.9668 | 1               |
|                           | 83.92   | 4        | 0        | 5.8744    | 1               |
| Cordless Telephone        | 131.98  | 2        | 0        | 35.6346   | 1               |
|                           | 15.92   | 4        | 0        | 7.4824    | 1               |
|                           | 52.29   | 9        | 0        | 16.2099   | 1               |
|                           | 91.99   | 1        | 0        | 3.6796    | 1               |
|                           | 1.624   | 2        | 0.8      | -4.466    | 1               |
|                           | 219.075 | 3        | 0.5      | -131.445  | 1               |
|                           | 25.92   | 5        | 0.2      | 9.396     | 1               |
|                           | 53.424  | 3        | 0.2      | 4.6746    | 1               |
|                           | 28.4    | 2        | 0        | 11.076    | 1               |
|                           | 287.92  | 8        | 0        | 138.2016  | 1               |

Figure 5

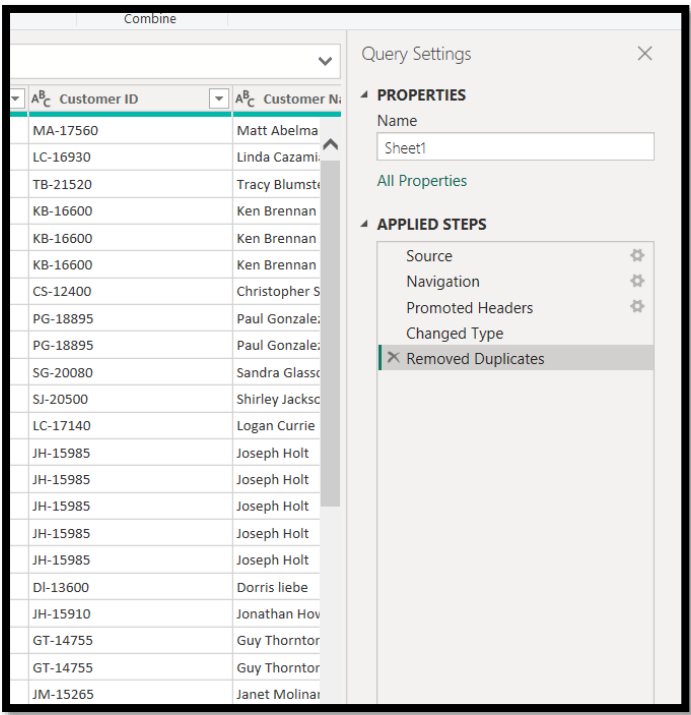


Figure 6

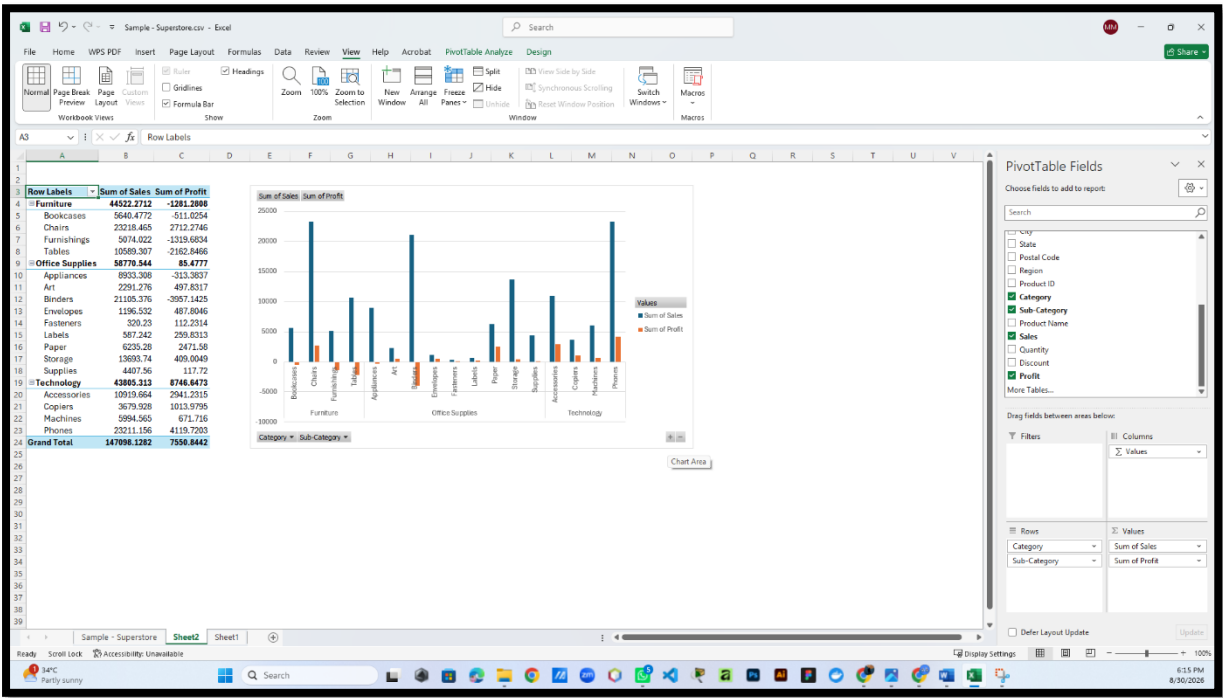


Figure 7

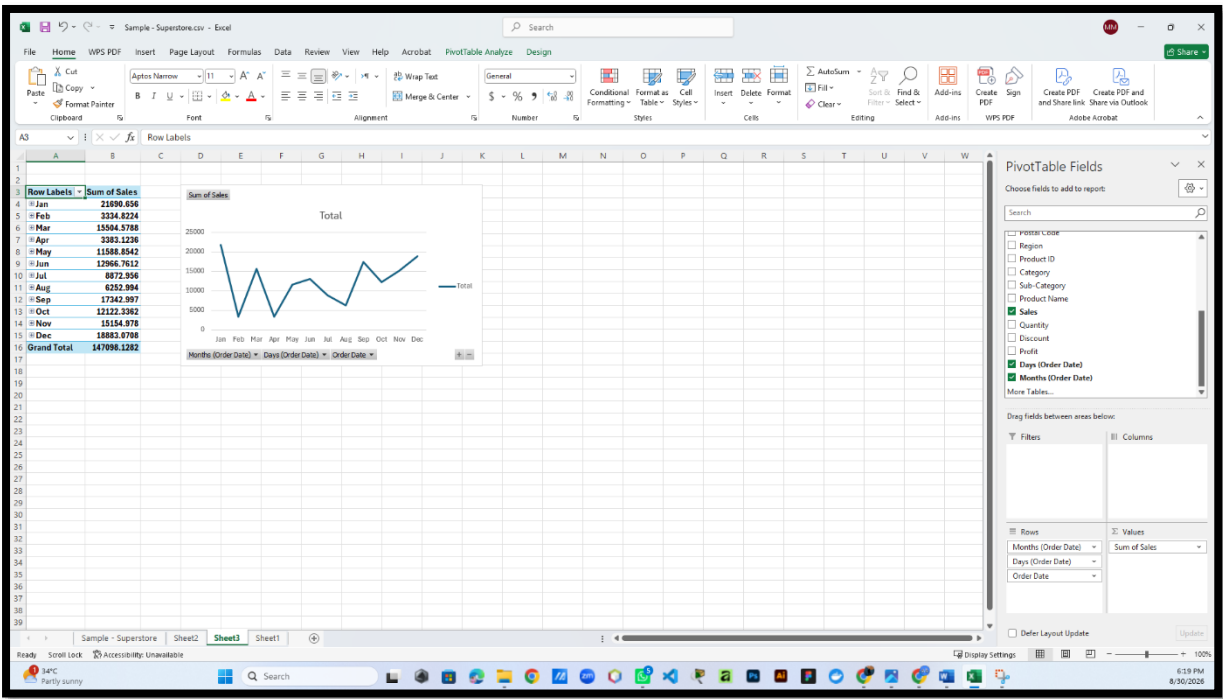


Figure 8

